# The Lemke Oliver–Soundararajan Bias and Ternary Constitutional Structure

Khayyam Wakil

ARC Institute of Knowware

Calgary, Alberta, Canada

the@knowware.institute

March 2026; revised November 2026

*This paper was not planned. On the Friday afternoon before submitting the Constitutional Forcing programme to peer review, the author read a popular account of the Lemke Oliver–Soundararajan prime digit bias. The first version of this paper, written within that afternoon and without calculation, proposed that the bias is a projection of the mod 3 structure of the primes. The present revision tests that proposal against recomputed data and against the Hardy–Littlewood singular series. It does not survive. What remains is an account of the genuine, but ordinary, part that the prime 3 plays inside the Lemke Oliver–Soundararajan explanation.*

**Abstract**

In 2016 Lemke Oliver and Soundararajan observed that consecutive primes are biased against repeating their residue modulo $q$, for instance their last decimal digit. In the same paper they explained the bias by a heuristic built on the Hardy–Littlewood prime $k$-tuples conjecture, with explicit lower-order terms whose leading part has size $\log\log x/\log x$. The first version of the present paper proposed a different, "ternary constitutional" mechanism: that the bias is the base-10 projection of the mod 3 constraint on primes, that the $p = 3$ factor of the singular series vanishes for gaps divisible by 3, and that the slow decay of the bias is the Siegel–Walfisz rate. We test these claims against a recomputation of the consecutive-prime transition counts for the first $10^8$ and $10^9$ primes. All three are false. The local factor at $p = 3$ is $3/2$, its maximum, for gaps divisible by 3 and $3/4$ otherwise; gap 6 is the most common gap; $\mathfrak{S}(2)$ is the smallest even-gap singular-series value, not the largest; the last digit of a prime carries no information about its residue mod 3; and the bias is about the same modulo 4 as modulo 3. A simple Hardy–Littlewood model with the correct $p = 3$ factor reproduces the base-10 transition matrix to 0.85 percentage points (RMS); the factor claimed in the first version gives 5.9 points and largely removes the diagonal suppression. We conclude that the Lemke Oliver–Soundararajan explanation accounts for the bias, and that the ternary-constitutional explanation, as originally stated, is not supported. The prime 3 enters only in the ordinary way, through the singular series, which weights gaps divisible by 3 twice as heavily as other gaps.

**Revision note (November 2026).** The main claim of the first version, that the bias is explained by ternary constitutional structure, is withdrawn, and the title has changed accordingly. The "observed" transition matrix, whose rows for the digits 3, 7 and 9 were wrong, has been replaced by a matrix recomputed for the first $10^8$ primes (the range of Lemke Oliver and Soundararajan's table) and for the first $10^9$ primes, and all four figures, which showed that matrix or the withdrawn mechanism, have been removed. Section 4 now states the singular-series facts correctly: the $p = 3$ factor is $3/2$ or $3/4$, never zero; $\mathfrak{S}(2)$ is the minimum even-gap value; and $\mathfrak{S}(10) \approx 1.76$. The minimum-gap ternary-class rule, the identification of the "snail's pace" with the Siegel–Walfisz rate, and the base-independence argument are withdrawn because the data refute them. The twin-prime count, the description of the parity barrier, the prime-gap bounds, the Pascadi reference and a misattributed quotation have been corrected, and two quotations that could not be traced to any source have been removed. The level $5/8$ for moduli $3^K$ is now described as a conjecture.

**Keywords:** prime digit bias, consecutive primes, Lemke Oliver–Soundararajan, Hardy–Littlewood singular series, mod 3 structure, twin primes, parity barrier
**MSC2020:** 11N05 (primary), 11N13, 11A41 (secondary)

# 1 Introduction

In 2016 Lemke Oliver and Soundararajan [LOS16] published a result that anyone with a computer can check. If the last digits of consecutive primes were independent, each of the four digits $\{1, 3, 7, 9\}$ would follow any other about 25% of the time. They do not. Among the first $10^8$ primes, a prime ending in 1 is followed by another prime ending in 1 only 18.5% of the time, and among the first $10^9$ primes a prime ending in 9 is followed by a prime ending in 1 about 65% more often than by another prime ending in 9 (Section 2). The same kind of bias is observed for other moduli (Section 6.1), and it is predicted for every modulus $q \geq 3$, not only for last digits (which are residues modulo 10). It fades as the primes grow, but slowly: "at a snail's pace", in the words of Quanta Magazine [Kla16].

Lemke Oliver and Soundararajan also explained the bias, in the same paper. From the Hardy–Littlewood prime $k$-tuples conjecture [HL23], together with uniformity assumptions, they derived a conjectural asymptotic formula for the number of consecutive primes in each pair of residue classes modulo $q$. Its second-order terms are explicit. The leading correction has size $\log\log x/\log x$ and depends only on $\varphi(q)$ and on whether the two classes coincide; this is why the bias appears for every modulus $q \geq 3$ and why it decays slowly. The argument is a heuristic, not a proof, since it rests on the $k$-tuples conjecture. But it is quantitative, and it fits the data well.

The first version of this paper proposed a different explanation: that the bias is "the base-10 projection of ternary constitutional pressure", that is, of the fact that every prime $p > 3$ lies in one of the classes $1, 2 \pmod 3$. It asserted that

(i) the $p = 3$ factor of the Hardy–Littlewood singular series vanishes for gaps divisible by 3;

(ii) gap 2 has the largest even-gap singular series and gap 10 a value below 1;

(iii) the ternary class of the smallest gap compatible with a digit transition governs whether that transition is enhanced or suppressed; and

(iv) the "snail's pace" is the Siegel–Walfisz rate at which primes become equidistributed modulo 3.

This revision checks each assertion against the singular series and against recomputed data. All four are false (Sections 4–6).

**Main result (revised, informal).** *The Lemke Oliver–Soundararajan bias is accounted for by their Hardy–Littlewood heuristic. The prime 3 enters that heuristic through its local factor,* $3/2$ *for gaps divisible by 3 and* $3/4$ *for other gaps, which favours gaps divisible by 3. This factor visibly shapes the base-10 transition matrix, but it is one local factor among many, and it is not what produces the diagonal suppression, which is an effect of gap length. The ternary-constitutional mechanism of the first version is not supported by the data.*

**Organisation.** Section 2 recalls the Lemke Oliver–Soundararajan findings, with recomputed data. Section 3 records the elementary mod 3 facts. Section 4 states the singular-series facts correctly and tests the first version's mechanism against the data. Section 5 discusses the rate of decay. Section 6 treats base dependence, the parity barrier, and the relation to the Constitutional Forcing programme. Section 7 concludes.

# 2 The Lemke Oliver–Soundararajan Findings

## 2.1 The Observation

For consecutive primes $p_n$ and $p_{n+1}$, consider the frequency with which the last digit of $p_n$ is followed by each possible last digit of $p_{n+1}$. The last digits of primes greater than 5 are drawn from $\{1, 3, 7, 9\}$, and if consecutive primes behaved independently each transition would occur with frequency about $1/4 = 25\%$. Lemke Oliver and Soundararajan [LOS16] tabulated these transitions for the first $10^8$ primes.

Table 1 gives the matrix recomputed with a segmented sieve for the first $10^8$ and the first $10^9$ primes. The sieve reproduces the standard values of $\pi(10^k)$ for $6 \le k \le 10$, of $p_{10^k}$ for $6 \le k \le 9$, and the twin-prime count $\pi_2(10^8) = 440{,}312$; for the first $10^8$ primes its counts for the row "ends in 1" agree with the published counts of [LOS16] to within one, the difference being an endpoint convention. (The code and raw output are kept with the programme's audit material, `audit/scripts/p08`.) Each row of the $10^8$ block contains about $2.5 \times 10^7$ transitions, and each row of the $10^9$ block about $2.5 \times 10^8$.

The headline finding is that *consecutive primes avoid repeating their last digit*: every diagonal entry is below 25%. Among the first $10^8$ primes the suppression is strongest for the digits 3 and 7 (17.8%) and weakest for 1 and 9 (18.5%). The ratio of $9 \to 1$ to $9 \to 9$ is 1.73 for the first $10^8$ primes and 1.65 for the first $10^9$; that is the "65% more often" of

Table 1: Last-digit transitions between consecutive primes: percentage of transitions from a prime $p_n$ ending in the row digit to a next prime $p_{n+1}$ ending in the column digit, recomputed for $n \le 10^8$ (so $p_n \le 2{,}038{,}074{,}743$, the range of the table in [LOS16]) and for $n \le 10^9$ ($p_n \le 22{,}801{,}763{,}489$).

| $p_n \to p_{n+1}$ | ends in 1 | ends in 3 | ends in 7 | ends in 9 |
|---|---|---|---|---|
| *First* $10^8$ *primes* | | | | |
| ends in 1 | 18.49 | 29.72 | 30.02 | 21.77 |
| ends in 3 | 24.04 | 17.77 | 28.17 | 30.01 |
| ends in 7 | 25.50 | 27.02 | 17.76 | 29.73 |
| ends in 9 | 31.97 | 25.49 | 24.05 | 18.49 |
| *First* $10^9$ *primes* | | | | |
| ends in 1 | 19.00 | 29.30 | 29.43 | 22.28 |
| ends in 3 | 24.25 | 18.45 | 27.87 | 29.43 |
| ends in 7 | 25.44 | 26.83 | 18.44 | 29.30 |
| ends in 9 | 31.31 | 25.43 | 24.26 | 19.00 |

Section 1. Relative to the uniform 25%, $9 \to 1$ is enhanced by a factor 1.28 (first $10^8$ primes).

The matrix is also symmetric under $(a \to b) \mapsto (-b \to -a) \pmod{10}$: for instance $3 \to 1$ is 24.04% and $9 \to 7$ is 24.05%. Corresponding entries agree to within 0.02 percentage points for the first $10^8$ primes. This is the symmetry $n \mapsto -n$, under which Hardy–Littlewood densities are invariant.

The first version of this paper printed a different matrix, described as approximate data for the first billion primes and attributed to [LOS16]. Only its first row was close to the data. Its rows for the digits 3, 7 and 9 were wrong by as much as 14 percentage points (for example $3 \to 1$ was given as 38% and $3 \to 9$ as 20%, against 24.04% and 30.01%), and they violated the symmetry just described. That matrix, and every conclusion drawn from it, has been withdrawn.

## 2.2 The Lemke Oliver–Soundararajan Explanation

Lemke Oliver and Soundararajan derived the bias from the Hardy–Littlewood prime $k$-tuples conjecture [HL23], which predicts the density of prime pairs $(p, p+h)$ through the singular series $\mathfrak{S}(h)$ (Section 4.2). Let $\pi(x; q, (a, b))$ count the primes $p_n \le x$ with $p_n \equiv a$ and $p_{n+1} \equiv b \pmod{q}$. Their conjecture [LOS16] reads

$$\pi(x; q, (a, b)) = \frac{\mathrm{li}(x)}{\varphi(q)^2} \left( 1 + c_1(q; a, b) \, \frac{\log\log x}{\log x} + c_2(q; a, b) \, \frac{1}{\log x} + O\big((\log x)^{-7/4}\big) \right),$$

with $c_1(q; a, b) = 1/2$ for $a \not\equiv b$ and $c_1(q; a, a) = 1/2 - \varphi(q)/2$. The coefficient $c_2(q; a, b)$ is explicit; it is computed from the Hardy–Littlewood singular series and depends on the

pair $(a, b)$. The leading correction is negative on the diagonal for every $q \geq 3$, which is the diagonal suppression; it depends only on $\varphi(q)$, which is why the effect looks alike for different moduli; and it decays like $\log\log x/\log x$, which is the slow decay. The finer structure of the off-diagonal entries comes from $c_2$. Lemke Oliver and Soundararajan studied the distribution of these biases further in [LOS20].

Their argument is a heuristic: it assumes the $k$-tuples conjecture in a uniform form. It is nevertheless quantitative and it agrees well with the data, and nothing in the present paper supplies an alternative to it.

# 3 The Ternary Constitutional Framework

## 3.1 Core Definitions

We work with remainder classes modulo 3. Every integer $n$ belongs to exactly one of three classes:

- $n \equiv 0 \pmod 3$: the **Forbidden Class** (divisible by 3);
- $n \equiv 1 \pmod 3$: **Constitutional Class I**;
- $n \equiv 2 \pmod 3$: **Constitutional Class II**.

**Definition 3.1** (Constitutional Constraint)**.** A prime $p > 3$ is *constitutionally constrained*: it must satisfy $p \equiv 1$ or $p \equiv 2 \pmod 3$. The Forbidden Class is closed to primes greater than 3.

This is elementary: any integer $n > 3$ with $n \equiv 0 \pmod 3$ is divisible by 3 and hence composite. Therefore all primes $p > 3$ lie in $\{p : p \equiv 1 \text{ or } 2 \pmod 3\}$.

## 3.2 The Twin Prime Constitutional Signature

The following fact is classical and elementary.

**Theorem 3.2** (Constitutional Signature)**.** *Let $(p, p+2)$ be a twin prime pair with $p > 3$. Then*

$$p \equiv 2 \pmod 3 \quad \textit{and} \quad p + 2 \equiv 1 \pmod 3.$$

*Proof.* Among any three consecutive integers, exactly one is divisible by 3.

- If $p \equiv 0 \pmod 3$: then $p = 3$ (the only such prime), excluded.
- If $p \equiv 1 \pmod 3$: then $p+2 \equiv 0 \pmod 3$, so $p+2 > 3$ is composite. Contradiction.
- If $p \equiv 2 \pmod 3$: then $p + 2 \equiv 1 \pmod 3$. Neither $p$ nor $p + 2$ is divisible by 3. Both may be prime. □

*Remark* 3.3. There are 3,424,506 twin prime pairs $(p, p+2)$ with $p \leq 10^9$. The pair $(3, 5)$ is excluded by the hypothesis $p > 3$; the remaining 3,424,505 pairs all have $p \equiv 2 \pmod 3$, as the theorem requires. The count is a check on the computation, not additional evidence for a two-line proof.

*Remark* 3.4. The same argument applies to any even gap $h$. If $3 \nmid h$, a pair of primes $(p, p+h)$ with $p > 3$ is possible for only one class of $p \bmod 3$ (namely $p \equiv 2$ if $h \equiv 2$, and $p \equiv 1$ if $h \equiv 1 \pmod 3$). If $3 \mid h$, both classes are possible. Gaps divisible by 3 are therefore *less* constrained modulo 3, not forbidden. This is the source of the $p = 3$ factor in the singular series (Section 4.2).

### 3.3 The Ternary Equilibrium

**Theorem 3.5** (Ternary Equilibrium; de la Vallée Poussin [dlVP96])**.** *Let $\pi_1(N)$ and $\pi_2(N)$ denote the number of primes $p \leq N$ with $p \equiv 1 \pmod 3$ and $p \equiv 2 \pmod 3$ respectively. Then*

$$\frac{\pi_1(N)}{\pi(N)} \to \frac{1}{2} \quad \text{and} \quad \frac{\pi_2(N)}{\pi(N)} \to \frac{1}{2} \quad \text{as } N \to \infty.$$

This is the case $q = 3$ of the prime number theorem for arithmetic progressions, proved by de la Vallée Poussin [dlVP96]. Dirichlet [Dir37] had proved earlier that each of the two classes contains infinitely many primes.

*Remark* 3.6 (Last digits carry no ternary information). Since $\gcd(10, 3) = 1$, the Chinese remainder theorem identifies the reduced residues modulo 30 with pairs (reduced residue mod 10, reduced residue mod 3). By the prime number theorem for arithmetic progressions modulo 30, each of the eight reduced classes mod 30 contains asymptotically 1/8 of the primes. Hence the last digit of a prime and its class modulo 3 are asymptotically independent: a prime ending in 3 or 9 is no more likely than any other to be $\equiv 1$ or $\equiv 2 \pmod 3$, and none is $\equiv 0$ except 3 itself. In particular, the last digit is not a "projection" of the class modulo 3.

## 4 The Hardy–Littlewood Explanation and the Role of the Prime 3

### 4.1 Minimum Gaps and the Withdrawn Mechanism

Each last-digit transition $(d_1 \to d_2)$ has a *minimum possible gap* $g_{\min}(d_1 \to d_2)$: the smallest positive even integer $g$ with $d_1 + g \equiv d_2 \pmod{10}$. Since all prime gaps for $p > 5$ are even, the possible gaps for the transition are exactly $g, g+10, g+20, \ldots$ The values are listed in Table 2; this part of the first version was correct.

**Withdrawn claim (first version)**

The first version claimed that the ternary class of $g_{\min}(d_1 \to d_2)$ governs whether the transition is enhanced, suppressed or "redistributed": gap-2 transitions enhanced because they carry the twin-prime signature, same-digit transitions suppressed because $\mathfrak{S}(10)$ is the smallest singular series, and gap-6 transitions redistributed because the $p = 3$ Euler factor vanishes for gaps divisible by 3. Each of these three statements is false (Sections 4.2–4.4), and the rule as a whole is refuted by the data.

Table 2: The sixteen last-digit transitions, grouped by minimum gap. "Observed" is the recomputed frequency for the first $10^8$ primes (Table 1). The last three columns are the crude Hardy–Littlewood model of Section 4.4 at $x = 2.04 \times 10^9$, with three choices of the $p = 3$ local factor: the correct one (3/2 if $3 \mid h$, 3/4 otherwise); the one claimed in the first version (0 if $3 \mid h$, 1/2 otherwise); and none (factor 1). All entries are percentages of the row.

| Transition | $g_{\min}(d_1 \to d_2)$ | $g$ mod 3 | Observed | Model, $p = 3$ factor | | |
|---|---|---|---|---|---|---|
| | | | | correct | first version | none |
| $9 \to 1$ | 2 | 2 | 31.97 | 31.15 | 27.25 | 29.69 |
| $1 \to 3$ | 2 | 2 | 29.72 | 30.33 | 31.46 | 30.71 |
| $7 \to 9$ | 2 | 2 | 29.73 | 29.51 | 28.47 | 29.14 |
| $3 \to 7$ | 4 | 1 | 28.17 | 26.91 | 33.74 | 29.33 |
| $7 \to 1$ | 4 | 1 | 25.50 | 24.96 | 31.58 | 27.29 |
| $9 \to 3$ | 4 | 1 | 25.49 | 26.36 | 30.23 | 27.81 |
| $1 \to 7$ | 6 | 0 | 30.02 | 28.31 | 19.45 | 25.34 |
| $3 \to 9$ | 6 | 0 | 30.01 | 29.70 | 18.80 | 25.84 |
| $7 \to 3$ | 6 | 0 | 27.02 | 27.55 | 17.60 | 24.05 |
| $3 \to 1$ | 8 | 2 | 24.04 | 24.00 | 23.59 | 23.85 |
| $9 \to 7$ | 8 | 2 | 24.05 | 23.50 | 21.13 | 22.61 |
| $1 \to 9$ | 8 | 2 | 21.77 | 22.88 | 24.39 | 23.39 |
| $1 \to 1$ | 10 | 1 | 18.49 | 18.48 | 24.70 | 20.57 |
| $3 \to 3$ | 10 | 1 | 17.77 | 19.39 | 23.88 | 20.98 |
| $7 \to 7$ | 10 | 1 | 17.76 | 17.98 | 22.35 | 19.52 |
| $9 \to 9$ | 10 | 1 | 18.49 | 18.99 | 21.39 | 19.89 |
| RMS difference from observed (percentage points) | | | | 0.85 | 5.94 | 2.36 |

## 4.2 The Singular Series and the Factor at $p = 3$

For even $h > 0$, the Hardy–Littlewood conjecture [HL23] predicts

$$\#\{p \le x : p \text{ and } p + h \text{ prime}\} \sim \mathfrak{S}(h)\,\frac{x}{(\log x)^2}, \qquad \mathfrak{S}(h) = \prod_p \frac{1 - \nu_p(h)/p}{(1 - 1/p)^2},$$

where $\nu_p(h)$ is the number of residue classes modulo $p$ occupied by $\{0, h\}$: $\nu_p(h) = 1$ if $p \mid h$ and $\nu_p(h) = 2$ otherwise. (These pairs need not be consecutive primes.) Evaluating the local factors gives the familiar form

$$\mathfrak{S}(h) = 2C_2 \prod_{\substack{p \mid h \\ p>2}} \frac{p-1}{p-2}, \qquad 2C_2 = 2\prod_{p>2} \frac{p(p-2)}{(p-1)^2} \approx 1.3203.$$

A local factor vanishes only when $\nu_p = p$, which for a pair happens only at $p = 2$ with $h$ odd; for even $h$ no factor vanishes.

**Correct statement**

At $p = 3$ the local factor is

$$\mathfrak{S}_3(h) = \frac{1 - \nu_3(h)/3}{(1-1/3)^2} = \frac{\chi_3(h)/2}{2/3} = \begin{cases} 3/2 & \text{if } 3 \mid h, \\ 3/4 & \text{if } 3 \nmid h, \end{cases}$$

where $\chi_3(h) = \#\{j \in \{1,2\} : j + h \not\equiv 0 \pmod 3\}$ counts the classes of $p \bmod 3$ for which $p + h$ is also prime to 3 (Remark 3.4). Thus $\chi_3(h)/2$ is the chance that $p + h$ is prime to 3 given that $p$ is, and $2/3$ is the same chance for a random integer. The factor is *largest* for gaps divisible by 3, and gaps divisible by 3 receive twice the weight of other gaps.

The first version wrote this factor as $(1 - 1/3)^{-1}(1 - \chi_3/\varphi(3))$ and concluded that it vanishes for $3 \mid h$ and equals $1/2$ otherwise. That formula inverts the role of $\chi_3$, which counts the favourable classes, and uses the normalisation $(1-1/3)^{-1}$ instead of $(1-1/3)^{-2}$; even on its own terms it gives $3/4$, not $1/2$, for $3 \nmid h$. It is withdrawn, together with the claim that gap-6 transitions "map primes to the Forbidden class".

The resulting values are

$$\mathfrak{S}(2) = \mathfrak{S}(4) = \mathfrak{S}(8) \approx 1.3203, \qquad \mathfrak{S}(6) = \mathfrak{S}(12) \approx 2.6406,$$
$$\mathfrak{S}(10) \approx 1.7604, \qquad \mathfrak{S}(14) \approx 1.5844, \qquad \mathfrak{S}(30) \approx 3.5209.$$

Since every factor $(p-1)/(p-2)$ exceeds 1, $\mathfrak{S}(2) = 2C_2$ is the *smallest* value over even $h$, attained exactly when $h$ is a power of 2. It is not the largest, as the first version stated, and $\mathfrak{S}(10) \approx 1.76$ is greater than 1, not less.

The data agree. Gap 6 is the most common gap between consecutive primes among the first $10^6$, $10^7$, $10^8$ and $10^9$ primes. Among the first $10^8$ primes it occurs 11,604,547 times, against 6,497,407 for gap 2 (a ratio of 1.79); among the first $10^9$ primes the counts are 105,002,853 and 58,047,180 (ratio 1.81). Gaps 2 and 4, which have the same singular series, are equally common: their ratio is 0.9998 among the first $10^8$ primes.

## 4.3 What the Transition Data Show

**The minimum-gap ternary class does not determine the effect.** Averaging the observed entries of Table 2 by minimum gap gives 30.47% for gap 2 ($g \equiv 2 \bmod 3$), 26.39% for gap 4 ($g \equiv 1$), 29.02% for gap 6 ($g \equiv 0$), 23.29% for gap 8 ($g \equiv 2$) and 18.13% for gap 10 ($g \equiv 1$). Gap 2 and gap 8 have the same ternary class, yet every gap-8 transition is below 25%. Gap 4 and gap 10 have the same class, yet gap 4 is above 25% and gap 10 is the most suppressed. Apart from gap 6, the averages decrease with the size of the gap. Gap 6 is raised by the $p = 3$ factor 3/2. This is what the Hardy–Littlewood heuristic predicts, and it is the opposite of "redistribution".

**Gap-6 transitions are enhanced, not redistributed.** The transitions $1 \to 7$ (30.02%) and $3 \to 9$ (30.01%) are the most frequent transitions in their rows. In particular $1 \to 7$ is slightly more frequent than the gap-2 transition $1 \to 3$ (29.72%; the corresponding values for the first $10^9$ primes are 29.43% and 29.30%).

**Gap-2 transitions are not dominated by twin primes.** The three transitions $9 \to 1$, $1 \to 3$ and $7 \to 9$ have the same set of possible gaps, $\{2, 12, 22, \ldots\}$. Among the first $10^8$ primes only 27.09% of the $9 \to 1$ transitions have gap 2, while 38.06% have gap 12, which is divisible by 3; for $1 \to 3$ the shares are 29.15% and 37.56%, and for $7 \to 9$ they are 29.16% and 37.55%. The claim that these transitions are enhanced because they "carry the twin-prime signature" is therefore withdrawn. The fact that $9 \to 1$ exceeds $1 \to 3$ is a matter of competition within a row: from the digit 1 the competitor $1 \to 7$ has minimum gap 6 and is boosted by the $p = 3$ factor, while from 9 the competitor $9 \to 3$ has minimum gap 4. In the model of Section 4.4, removing the $p = 3$ factor reverses the order of the two entries (Table 2).

**The diagonal suppression is a gap-length effect.** Every same-digit gap is a multiple of 10, so it receives the extra $p = 5$ factor 4/3; the singular series *favours* the diagonal. Same-digit transitions are suppressed all the same, because they need a gap of at least 10, and it is less likely that no prime occurs in a long interval. In the Lemke Oliver–Soundararajan conjecture this is the negative first-order term $c_1(q; a, a) \log\log x / \log x$ (Section 2). The suppression survives when all ternary information is removed: with the $p = 3$ factor set to 1, the model still gives diagonal entries between 19.5% and 21.0%. The same holds modulo 3, where the singular series favours the diagonal even more strongly: same-class gaps are multiples of 6 and receive the *maximal* $p = 3$ factor, yet the frequency of a repeated class modulo 3 is 44.67% for the first $10^8$ primes and 45.15% for the first $10^9$, below the uniform 50%.

## 4.4 A Crude Hardy–Littlewood Model

To compare the three choices of $p = 3$ factor directly, consider the model

$$P(d_1 \to d_2) \;\propto \sum_{\substack{h>0 \\ h \equiv d_2 - d_1 \ (10)}} \mathfrak{S}'(h)\, e^{-h/\log x},$$

normalised over each row, where $\mathfrak{S}'(h)$ is the product of the local factors of $\mathfrak{S}(h)$ over primes $p \nmid 10$ (the factors at 2 and 5 are fixed by conditioning on residues modulo 10), and $e^{-h/\log x}$ is a Cramér-type probability that no prime lies in between. We take $x = 2.04 \times 10^9$, the size of the $10^8$-th prime, truncate the product at $p < 20{,}000$ and the sum at $h < 40 \log x$. This is not the Lemke Oliver–Soundararajan conjecture, only a simple model built from the same ingredients; the code is `model.py` in the audit material.

The results are in Table 2. With the correct $p = 3$ factor the model reproduces the observed matrix to an RMS error of 0.85 percentage points. It is crude: its largest error is 1.7 points (for $1 \to 7$, 28.31% against 30.02%), and it places $1 \to 7$ below $1 \to 3$, whereas the data place it slightly above. With no $p = 3$ factor the RMS error rises to 2.36 points. With the factor claimed in the first version it is 5.94 points: that factor predicts $1 \to 7 = 19.45\%$ against an observed 30.02%, and it raises the diagonal entries to between 21.39% and 24.70% (observed: 17.76% to 18.49%), largely removing the diagonal suppression that the first version set out to explain.

*Remark* 4.1. The model and the Lemke Oliver–Soundararajan conjecture both rest on the Hardy–Littlewood conjecture, and neither is a proof. What the comparison shows is that the prime 3 matters, through its ordinary local factor 3/2 or 3/4, and that the mechanism proposed in the first version (a vanishing factor and "redistribution") is contradicted by the data. No further "mechanistic layer" beneath the singular series is needed to account for the matrix, and the first version's description of the singular series as "the quantitative expression of ternary constitutional structure" is withdrawn: the factor at 3 is one local factor among the factors at all primes.

# 5 The Snail's Pace: The Rate of Decay

The bias fades as the primes grow, but slowly. The phrase "at a snail's pace" is Quanta Magazine's [Kla16], not Lemke Oliver and Soundararajan's. Their conjecture gives the rate explicitly: the leading correction is $c_1(q; a, b) \log\log x / \log x$ (Section 2), which decays even more slowly than $1/\log x$.

**Withdrawn claim (first version)**

The first version identified the snail's pace with the Siegel–Walfisz rate of equidistribution of primes modulo 3, $|\pi_1(N) - \pi_2(N)| = O(N \exp(-c\sqrt{\log N}))$, and stated that the digit bias diminishes "at exactly the rate" of that bound. The identification, the corresponding proposition, and the figure that illustrated it are withdrawn.

The identification fails for four reasons.

(a) The bias concerns *pairs* of consecutive primes. It is present even though single primes are very well distributed among the residue classes, so a bound for $|\pi_1(N) - \pi_2(N)|$ cannot be its rate. In any case the last digit carries no information about the class modulo 3 (Remark 3.6).

(b) The bound $N\exp(-c\sqrt{\log N})$ is an upper bound for the error, coming from the classical zero-free region for the fixed modulus 3; the Siegel–Walfisz theorem [Wal36] makes such a bound uniform for moduli $q \le (\log N)^A$. It is not the true size of the error, and "at exactly the rate" of an $O$-bound has no meaning.

(c) The true size is much smaller. The recomputed differences $\pi(x;3,2) - \pi(x;3,1)$ are 190, 420, 2107 and 6264 at $x = 10^7, 10^8, 10^9, 10^{10}$, while $\sqrt{x}/\log x \approx 196$, 543, 1526, 4343. This is the size of Chebyshev's bias: under the generalised Riemann hypothesis and a linear-independence hypothesis, $(\log x/\sqrt{x})(\pi(x;3,2) - \pi(x;3,1))$ has a limiting logarithmic distribution [RS94]. The relative imbalance $(\pi(x;3,2) - \pi(x;3,1))/\pi(x)$ is $4.1 \times 10^{-5}$ at $x = 10^9$, whereas the diagonal entries of Table 1 for the first $10^8$ primes lie on average 6.9 percentage points below 25%.

(d) The two quantities do not decay together. From $x = 10^7$ to $x = 10^{10}$ the relative mod-3 imbalance falls by a factor of about 21, from $2.9 \times 10^{-4}$ to $1.4 \times 10^{-5}$. Over a comparable range (the first $10^6$ primes, up to $1.5 \times 10^7$, to the first $10^9$ primes, up to $2.3 \times 10^{10}$) the mean diagonal deficit of the base-10 matrix falls only from 8.50 to 6.28 percentage points, a factor of 1.35.

Finally, $\exp(-c\sqrt{\log N})$ decays *faster* than any fixed power of $1/\log N$, and hence faster than $\log\log N/\log N$; a quantity that decays this fast cannot be the slow decay that Lemke Oliver and Soundararajan derived. The first version's Figure 3, which asserted the opposite and plotted a hypothetical curve as "confirmation", has been removed.

# 6 Other Moduli, Parity, and the Programme

## 6.1 Other Moduli

**Withdrawn claim (first version)**

The first version argued that the bias is base-independent because "the mechanism operates in mod-3 arithmetic": in every base the prime last digits map to ternary classes, and the ternary class of the minimum gap determines the transition structure. That argument, and the figure that illustrated it for bases 6, 10 and 16, are withdrawn.

The argument fails on its own terms. When $\gcd(b,3) = 1$, as for $b = 10$ and $b = 16$, the last digit in base $b$ carries no information about the class modulo 3 (Remark 3.6); for instance, primes ending in 3 or 9 in base 10 are not divisible by 3, and the figure's count of "Forbidden-class prime terminal digits" contradicted Definition 3.1. When $3 \mid b$, as for $b = 6$, the last digit of a prime $p > 3$ is determined by $p$ mod 6, that is, by $p$ mod 3; the same-digit gaps are then multiples of 6 and receive the *maximal* $p = 3$ factor, yet the diagonal is suppressed (Section 4.3).

The correct statement is the one of Lemke Oliver and Soundararajan: their conjecture is formulated for every modulus $q$, and its first-order term depends only on $\varphi(q)$ and on whether $a \equiv b$. The data fit this. Modulo 4, where there is no ternary digit structure at

all, the frequency of a repeated class is 45.04% for the first $10^8$ primes and 45.48% for the first $10^9$, close to the values 44.67% and 45.15% modulo 3; both moduli have $\varphi(q) = 2$.

## 6.2 The Parity Barrier

Selberg's parity problem [Sel49] is the observation that classical sieve methods cannot distinguish integers with an even number of prime factors from integers with an odd number; in other words, they cannot detect the parity of $\Omega(n)$, the sign of the Liouville function. It has nothing to do with working "modulo 2". The first version described it as a statement about "sieve methods operating modulo 2" and cited a different paper of Selberg's, on the elementary proof of the prime number theorem for arithmetic progressions; both are corrected here.

For bounded gaps between primes the history is as follows. Zhang [Zha14] proved that $\liminf(p_{n+1} - p_n) \le 7 \times 10^7$. Maynard [May15] obtained 600, and 12 under the Elliott–Halberstam conjecture. The Polymath8b project [Pol14] obtained 246 unconditionally, using the Bombieri–Vinogradov level of distribution 1/2 [Bom65, Vin65], and 6 under the generalised Elliott–Halberstam conjecture. The value 246 is therefore set by the level of distribution, not by the parity problem. It is the parity problem that prevents these sieve methods from going below 6 even under the generalised Elliott–Halberstam conjecture, and so from reaching gap 2.

The first version also gave a table of "ternary signatures" of gaps and claimed that gap 2 is the only even gap avoiding the Forbidden class, so that a ternary method could isolate twin primes where sieves fail. That claim is withdrawn. The table silently fixed $p \equiv 2 \pmod 3$; for $p \equiv 1$, the pair $(p, p+4)$ avoids the Forbidden class, as does $(p, p+8)$ for $p \equiv 2$. The correct statement is Remark 3.4: each gap not divisible by 3 allows exactly one class of $p$ mod 3, and each gap divisible by 3 allows both. Table 3 lists the consequences for the first six even gaps.

Table 3: Small even gaps: the classes of $p$ mod 3 for which $(p, p+h)$ can be a pair of primes with $p > 3$, the local factor at $p = 3$, the singular series, and the number of times $h$ occurs as a gap between consecutive primes among the first $10^8$ primes.

| $h$ | $h$ mod 3 | classes of $p$ mod 3 | $\mathfrak{S}_3(h)$ | $\mathfrak{S}(h)$ | consecutive gaps |
|---|---|---|---|---|---|
| 2 | 2 | 2 | 3/4 | 1.3203 | 6,497,407 |
| 4 | 1 | 1 | 3/4 | 1.3203 | 6,496,371 |
| 6 | 0 | 1, 2 | 3/2 | 2.6406 | 11,604,547 |
| 8 | 2 | 2 | 3/4 | 1.3203 | 5,157,442 |
| 10 | 1 | 1 | 3/4 | 1.7604 | 6,682,808 |
| 12 | 0 | 1, 2 | 3/2 | 2.6406 | 8,621,685 |

In particular cousin primes $(p, p+4)$ are as common as twin primes, as $\mathfrak{S}(4) = \mathfrak{S}(2)$ predicts, and gap 6 is the most common gap. The condition $p \equiv 2 \pmod 3$ for twin

primes is a local condition of exactly the kind that every sieve, and the singular series, already take into account. It does not bear on the parity problem.

### 6.3 Relation to the Constitutional Forcing Programme

This paper is p08 of the Constitutional Forcing programme [Wakil26], which consists of nine papers, p01–p09. The first version described the programme as having established a level of distribution $\theta_W = 5/8$ for primes over cascade moduli $q = 3^K$. That level is a conjecture (p04 [Wakil26d]). For prime-power moduli the known results are well below 5/8, and a level of 5/8 for each individual modulus $3^K$ would go beyond what the generalised Riemann hypothesis gives. The value 5/8 also appears in Pascadi's work [Pas25] on smooth numbers and on primes with triply-well-factorable weights, for general moduli; that result does not concern moduli $3^K$, and the agreement of the two numbers is a coincidence, not evidence.

The digit bias is independent of all of this. Its explanation requires the Hardy–Littlewood heuristic and nothing from the programme. The first version's claims that the digit bias and the cascade-moduli level are "projections of the same underlying structure", and that the explanation needs only the elementary mod 3 constraint and the Siegel–Walfisz theorem, are withdrawn. The second claim was in any case inconsistent with the first version's own remark that the explanation inherits the assumptions of the Hardy–Littlewood conjecture.

## 7 Conclusion

The Lemke Oliver–Soundararajan bias is easy to check: Soundararajan told Nature [Lam16] that everyone they told about it ended up writing a program to check it for themselves. It was explained, heuristically but quantitatively, in the paper that announced it [LOS16]. This paper tested a different explanation, ternary constitutional structure, against recomputed data for the first $10^8$ and $10^9$ primes. The explanation does not survive:

(i) The diagonal suppression is a gap-length effect, the negative first-order term of the Lemke Oliver–Soundararajan conjecture. It is not a singular-series penalty: $\mathfrak{S}(10) \approx 1.76$ exceeds $\mathfrak{S}(2) \approx 1.32$, which is the smallest even-gap value, and the same-class gaps modulo 3 receive the maximal $p = 3$ factor and are still suppressed.

(ii) The slow decay is the $\log\log x/\log x$ term of that conjecture. It is not the Siegel–Walfisz rate; the mod-3 imbalance of single primes is of order $\sqrt{x}/\log x$, far smaller than the digit bias, and it decays far faster.

(iii) The bias is not a base-10 projection of mod-3 structure: last digits carry no information about the class mod 3, and the bias modulo 4 is about the same as modulo 3.

(iv) The rule that the ternary class of the minimum gap governs each transition is contradicted by the data (gap 2 against gap 8, gap 4 against gap 10), and the $p = 3$

factor does not vanish for gaps divisible by 3: it is 3/2, its maximum.

What the prime 3 does contribute is its ordinary local factor in the singular series, 3/2 for gaps divisible by 3 and 3/4 otherwise. This is why 6 is the most common gap between consecutive primes in the computed range. The factor also boosts the transitions $1 \to 7$, $3 \to 9$ and $7 \to 3$ (in the model of Section 4.4, removing it lowers $1 \to 7$ from 28.31% to 25.34%), and in that model it accounts for $9 \to 1$ exceeding $1 \to 3$. This effect is part of the Lemke Oliver–Soundararajan explanation, not an alternative to it. The elementary fact that twin primes $p > 3$ satisfy $p \equiv 2 \pmod 3$ (Theorem 3.2) is true, but it explains none of the bias.

The conclusion of the first version is therefore reversed. The Lemke Oliver–Soundararajan explanation accounts for the bias, and the ternary-constitutional explanation, as originally stated, is not supported.

# References


[LOS16] R. J. Lemke Oliver and K. Soundararajan, Unexpected biases in the distribution of consecutive primes, *Proc. Natl. Acad. Sci. USA*, 113(31):E4446–E4454, 2016. arXiv:1603.03720 [math.NT].

[LOS20] R. J. Lemke Oliver and K. Soundararajan, The distribution of consecutive prime biases and sums of sawtooth random variables, *Math. Proc. Cambridge Philos. Soc.*, 168(1):149–169, 2020. arXiv:1709.06168 [math.NT].

[HL23] G. H. Hardy and J. E. Littlewood, Some problems of 'Partitio numerorum'; III: on the expression of a number as a sum of primes, *Acta Math.*, 44:1–70, 1923.

[Dir37] P. G. L. Dirichlet, Beweis des Satzes, dass jede unbegrenzte arithmetische Progression, deren erstes Glied und Differenz ganze Zahlen ohne gemeinschaftlichen Factor sind, unendlich viele Primzahlen enthält, *Abhandlungen der Königlich Preussischen Akademie der Wissenschaften*, pages 45–81, 1837.

[dlVP96] C.-J. de la Vallée Poussin, Recherches analytiques sur la théorie des nombres premiers, *Ann. Soc. Sci. Bruxelles*, 20, 1896.

[Wal36] A. Walfisz, Zur additiven Zahlentheorie. II, *Math. Z.*, 40:592–607, 1936.

[RS94] M. Rubinstein and P. Sarnak, Chebyshev's bias, *Experiment. Math.*, 3(3):173–197, 1994.

[Bom65] E. Bombieri, On the large sieve, *Mathematika*, 12:201–225, 1965.

[Vin65] A. I. Vinogradov, The density hypothesis for Dirichlet $L$-series, *Izv. Akad. Nauk SSSR Ser. Mat.*, 29, 1965.

[Sel49] A. Selberg, On elementary methods in prime number theory and their limitations, In *Den 11te Skandinaviske Matematikerkongress, Trondheim, 1949*, Johan Grundt Tanums Forlag, Oslo, 1952.

[Zha14] Y. Zhang, Bounded gaps between primes, *Ann. of Math.*, 179(3):1121–1174, 2014.

[May15] J. Maynard, Small gaps between primes, *Ann. of Math.*, 181(1):383–413, 2015.

[Pol14] D. H. J. Polymath, Variants of the Selberg sieve, and bounded intervals containing many primes, *Res. Math. Sci.*, 1, Art. 12, 2014.

[Pas25] A. Pascadi, On the exponents of distribution of primes and smooth numbers, arXiv:2505.00653, 2025.

[Wakil26] K. Wakil, Constitutional Forcing, ARC Institute of Knowware, preprint, 2026.

[Wakil26d] K. Wakil, Ternary Forcing on the Hexagonal Lattice: Eisenstein Integers, Cascade Moduli, and a Conjectured Level of Distribution 5/8, ARC Institute of Knowware, preprint, 2026.

[Kla16] E. Klarreich, Mathematicians discover prime conspiracy, *Quanta Magazine*, 13 March 2016.

[Lam16] E. Lamb, Peculiar pattern found in "random" prime numbers, *Nature* (news), 14 March 2016; reprinted in *Scientific American*.